Routing Dinamis dengan 4 Router

Metode RIPv2(Routing Information Protocol)dan OSPF(Open Shortest Path First)

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A.Tujuan Praktikum

1. Membangun topologi jaringan dengan 4 Router,4 pc melalui cisco packet tracer.

2. Mengonfigurasi alamat IP pada interface router dan PC.

3. Mengonfigurasi routing dinamis RIPv2 dan OSPF pada 4 router.

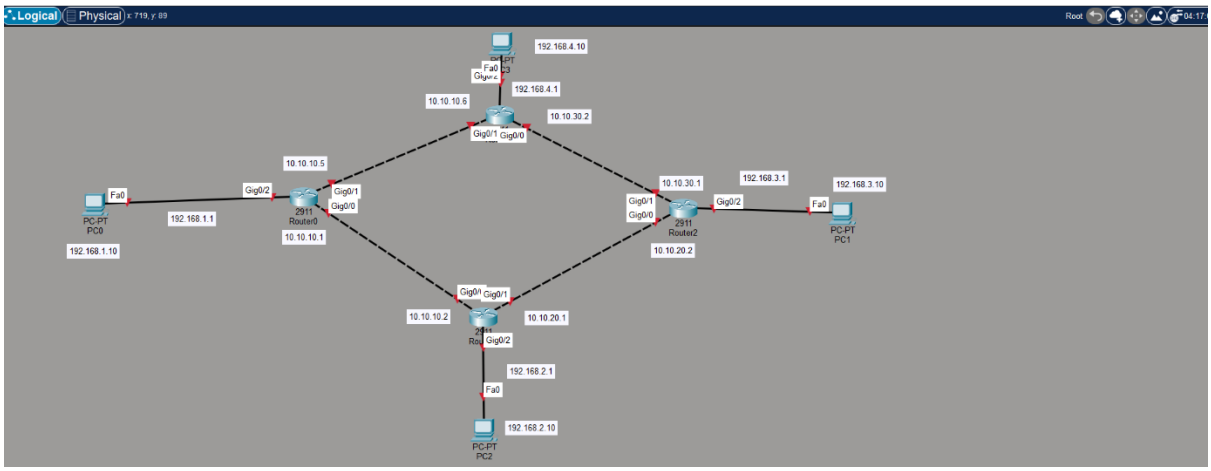
4. Memverifikasi konektivitas.

B.Alat dan Bahan

No	Perangkat	Jumlah	Keterangan
1	Router cisco 2911	4	R1,R2,R3,R4
2	PC(Personal Computer)	4	PC0,PC1,PC2,PC3
3	Kabel copper straight-through	4	PC ke Router
4	Kabel copper cross-over	4	Router ke Router

D.Topologi 4 Router

Buatlah desain Topologi 4 Router seperti pada Gambar yang saya cantumkan.Yang dimana 4 router itu saling terhubung satu sama lain menggunakan kabel cross-over dan pc ke router terhubung melalui kabel straight-through.



E.Konfigurasi Interface Router

R1

Router>en

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#int g0/2

Router(config-if)#ip add 192.168.1.1 255.255.255.0

Router(config-if)#no sh

Router(config)#int g0/0

Router(config-if)#ip add 10.10.10.1 255.255.255.252

```
Router(config-if)#no sh

Router(config)#int g0/1
Router(config-if)#ip add 10.10.10.5 255.255.255.252
Router(config-if)#no sh
```

R2

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/2
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#no sh

Router(config)#int g0/0
Router(config-if)#ip add 10.10.10.2 255.255.255.252
Router(config-if)#no sh

Router(config)#int g0/1
Router(config-if)#ip add 10.10.20.1 255.255.255.252
Router(config-if)#no sh
```

R3

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/2
Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config-if)#no sh

Router(config)#int g0/0
Router(config-if)#ip add 10.10.20.2 255.255.255.252
Router(config-if)#no sh

Router(config)#int g0/1
Router(config-if)#ip add 10.10.30.1 255.255.255.252
Router(config-if)#no sh
```

R4

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/2
Router(config-if)#ip add 192.168.4.1 255.255.255.0
Router(config-if)#no sh

Router(config)#int g0/1
Router(config-if)#ip add 10.10.10.6 255.255.255.252
```

Router(config-if)#no sh

Router(config)#int g0/0
Router(config-if)#ip add 10.10.30.2 255.255.255.252
Router(config-if)#no sh

F.Command Routing RIPv2

R1

Router(config)#router rip
Router(config-router)#v 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#passive-interface g0/2
Router(config-router)#end

R2

Router(config)#router rip
Router(config-router)#v 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#passive-interface g0/2

Router(config-router)#end

R3

Router(config)#router rip

Router(config-router)#v 2

Router(config-router)#no auto-summary

Router(config-router)#network 192.168.3.0

Router(config-router)#network 10.0.0.0

Router(config-router)#passive-interface g0/2

R4

Router(config)#router rip

Router(config-router)#v 2

Router(config-router)#no auto-summary

Router(config-router)#network 192.168.4.0

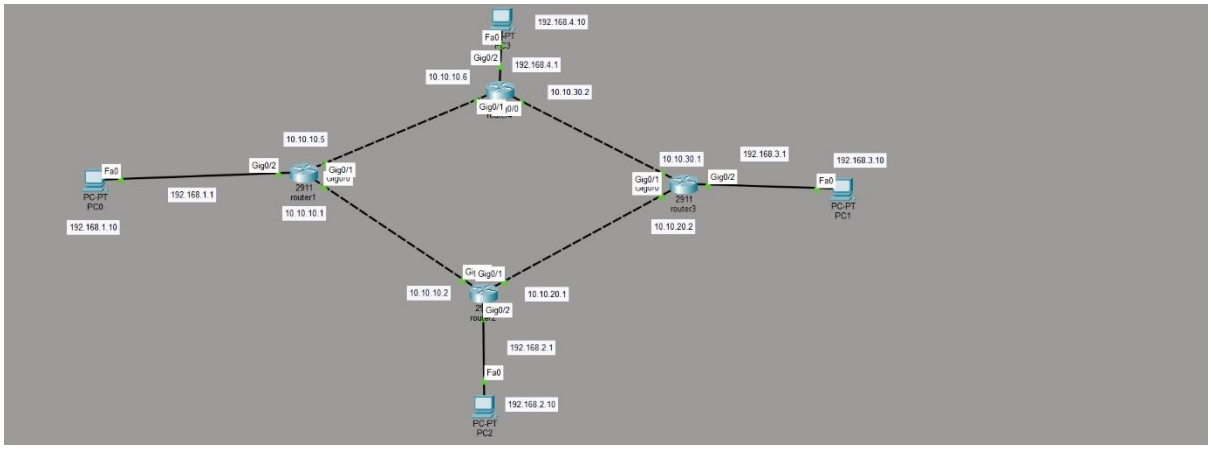
Router(config-router)#network 10.0.0.0

Router(config-router)#passive-interface g0/2

Noted: Command Passive-interface untuk tidak memasukkan interface yang tidak ingin di masukkan dalam routing dan hanya sebagai interface biasa.Harus menyesuaikan dengan interface port yang ingin anda pakai,semisal saya mengkoneksikan router ke pc melalui g0/2 .

Command No auto-summary untuk menonaktifkan peringkasan rute otomatis (automatic summarization) ke dalam alamat jaringan *classful* (kelas A, B, atau C), sehingga router dapat mengiklankan subnet secara spesifik (*classless*). [1,2,3,4].

G.Hasil konfigurasi dan successful mengirim pesan



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	router1	router2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	router1	router3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC3	PC2	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	router3	ICMP		0.000	N	3	(edit)	(delete)

Jika menggunakan metode mengirim surat/PDU

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Jika menggunakan metode Command Prompt

H.Command Routing OSPF

R1

```
Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.10.4 0.0.0.3 area 0
Router(config-router)#passive-interface g0/2
```

R2

```
Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#network 10.10.20.0 0.0.0.3 area 0
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#passive-interface g0/2
```

R3

```
Router(config)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 10.10.20.0 0.0.0.3 area 0
Router(config-router)#passive-interface g0/2
```

R4

```
Router(config)#router ospf 1
```

```
Router(config-router)#router-id 4.4.4.4
Router(config-router)#network 192.168.4.0 0.0.0.255 area 0
Router(config-router)#network 10.10.10.4 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#passive-interface g0/2
```